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Report on the Coordination of Volunteer Tutors in Educational Communities

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Declaration

We, Jose Luis Leudo Mosquera, David Santiago Rojas Hernández, Adrián Mauricio Hernández Ortiz, and Johan Sebastian Candia Marín, students of the bachelor of Science in Computer Science, Universidad Distrital Francisco José de Caldas, Bogotá D.C., Colombia, confirm that this is our own work and that figures, tables, equations, code snippets, and illustrations in this report are original and have not been taken from any other person's work, except where the works of others have been explicitly acknowledged, quoted, and referenced. We understand that failing to do so would be considered a case of academic misconduct.

The simulation models described in this report (`simulation_1_discrete_event.py` and `simulation_2_cellular_automata.py`) were developed by the team and are available in our public GitHub repository. The survey data was collected directly from volunteer respondents at Universidad Distrital Francisco José de Caldas. All figures and tables were produced by the authors from original data and simulation outputs.

This report was prepared as the final deliverable of the Systems Analysis and Design course, under the guidance of Professor Carlos Andrés Sierra, M.Sc., during the 2026-1 academic semester.

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Abstract

School dropout in vulnerable urban communities is a persistent challenge driven by multiple factors, one of which is the lack of timely and personalized academic support. In Bogotá D.C., there is a meaningful population of university students willing to volunteer as tutors, yet the coordination between these volunteers and the educational institutions that need them is currently manual, fragmented, and hard to scale.

This report describes the full systems engineering lifecycle applied to the design, implementation, and computational validation of a Community Volunteer Tutor Coordination System. The system's core is a weighted compatibility algorithm, $C(v, p)$, whose four parameters were derived from an empirical survey of 50 potential volunteers at Universidad Distrital Francisco José de Caldas and validated through two complementary simulation models: a Discrete-Event Simulation of the assignment pipeline, and a Cellular Automata model of volunteer engagement dynamics across the university community.

Simulation results confirmed that the algorithm achieves mean compatibility scores between 0.860 and 0.887 across all tested scenarios, achieves 100% project coverage with a minimum pool of 30 active volunteers, and produces an 18% relative increase in community engagement as an emergent property of the volunteer social network. A previously unidentified risk, volunteer burnout contagion, was discovered through the cellular automata model, leading to a new system requirement for load monitoring and geographic load distribution.

Keywords: volunteer coordination, compatibility algorithm, systems analysis, vulnerable communities, educational support, Bogotá.

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Glossary

Burnout contagion: The social process by which volunteer fatigue and disengagement spreads through proximity networks. In the cellular automata model, a volunteer's burnout risk increases as more burned-out volunteers appear in their immediate social neighborhood.

Cellular Automata (CA): A computational model consisting of a grid of cells that evolve through discrete time steps according to local transition rules. Used in this project to model volunteer engagement dynamics across the university campus.

Compatibility score $C(v, p)$: A scalar value between 0 and 1 computed for each volunteer-project pair by the matching algorithm. Scores above 0.80 are considered acceptable for assignment.

Discrete-Event Simulation (DES): A simulation paradigm in which the system state changes only at discrete points in time, triggered by specific events. Used in this project to model the volunteer assignment pipeline.

Emergent behavior: System-level behavior that arises from the interactions of individual components and cannot be predicted by analyzing any single component in isolation.

Moore neighborhood: In a cellular automata grid, the eight cells surrounding a given cell (horizontally, vertically, and diagonally). Used to define local social influence in the CA model.

Pool: The set of active volunteers available for assignment at a given point in time.

Quality paradox under scarcity: The observed phenomenon in the stress simulation scenario where a smaller volunteer pool produces higher mean compatibility scores, because the algorithm concentrates assignments on the best available matches rather than distributing to marginally qualified volunteers.

Sociotechnical system: A system composed of both social elements (people, organizations, relationships) and technical elements (software, hardware, processes) that interact to produce system-level outcomes.

Tipping point: In the CA model, the threshold of approximately 15% neighborhood activation above which volunteer engagement accelerates non-linearly through self-sustaining recruitment cascades.

Chapter 1

Introduction

School dropout in communities with limited economic resources is not simply an academic problem. It reflects a broader gap between the educational support that students need and what their schools can realistically provide. In Bogotá D.C., many public institutions serving low-income neighborhoods rely on external volunteers for tutoring support, yet there is no structured system to connect these volunteers with the right institutions at the right time.

What makes this problem particularly difficult is that it is not a shortage of willing people. Our survey of 50 students at Universidad Distrital Francisco José de Caldas showed that approximately 78% expressed interest in participating in volunteer tutoring programs. The bottleneck is coordination. Volunteers do not know which institutions need them, institutions do not know which volunteers are available and qualified for their specific subject needs, and the matching process, when it happens at all, is done manually through messages and informal networks.

This report documents the work we carried out over four workshops to analyze this system, design a solution, manage its risks, and validate the design through computational simulation. The result is not just a platform, but a full systems engineering study of a sociotechnical problem: how to make volunteer academic support reliable, traceable, and scalable in a real urban context.

The report is organized as follows. Chapter 2 reviews relevant literature and prior approaches. Chapter 3 states the objectives and scope. Chapter 4 covers assumptions and limitations. Chapter 5 describes the methodology and system design in detail. Chapter 6 presents the simulation results. Chapter 7 discusses findings and their implications. Chapter 8 concludes and proposes future work.

1.1 Objectives

The primary objective of this project is to design, document, and computationally validate a community volunteer tutor coordination system that can replace the current informal, manual matching process with a structured, traceable, and scalable alternative.

More specifically, the project sought to achieve the following:

To characterize the current volunteer tutoring system in Bogotá D.C. through primary data collection and stakeholder analysis, identifying the structural causes of its inefficiency. To design a weighted compatibility algorithm whose parameters are directly grounded in empirical survey data rather than assumptions. To propose a system architecture aligned with recognized quality and risk management standards. To validate the design through simulation before any real-world deployment, identifying both expected outcomes and emergent risks. And finally, to produce a set of actionable recommendations for system launch and operation.

1.2 Scope

This project covers the analysis, design, and simulation of the volunteer coordination system. It does not include the full software implementation or deployment of the platform, which is planned for a subsequent phase.

The system under study includes the following: the population of potential volunteer tutors at Universidad Distrital Francisco José de Caldas, the educational institutions in vulnerable communities of Bogotá D.C. that require academic support, the coordination processes that connect these two groups, and the information flows and decisions involved in assigning a tutor to a student project.

The scope explicitly excludes the internal pedagogical methods used during tutoring sessions, the academic content of the tutoring programs, and any processes internal to participating educational institutions beyond their role as requesters of tutoring support.

1.3 Assumptions

Several assumptions were necessary to carry out the design and simulation work. We assumed that the survey sample of 50 volunteers is reasonably representative of the broader volunteer population at the university, given that it was distributed through academic groups and student networks and captured a range of study areas and availability profiles. We assumed that the weight structure of the compatibility algorithm remains stable in the short term and can be recalibrated using real post-session satisfaction data once a pilot is running. We assumed that the session cancellation rate of approximately 15%, drawn from initial observations during data collection, provides a realistic baseline for stress testing the system. And we assumed that Google Sheets, used as a temporary database in the prototype, is sufficient for a pilot of up to 100 volunteers before migration to a more robust data layer is required.

1.4 Limitations

The most significant limitation of this study is that the compatibility algorithm has not yet been validated with real assignment and satisfaction data from a deployed system. The weights and thresholds are grounded in survey evidence, but they may need adjustment once actual tutor-student sessions take place and feedback is collected.

The simulation models, while computationally validated and reproducible, are inherently abstractions. The discrete-event simulation assumes a stable 12-week horizon and simplified churn dynamics, and the cellular automata model represents the university community as a uniform 30x30 grid that does not capture the actual spatial and social topology of the campus. The burnout contagion rule, in particular, is a plausible model but has not been empirically calibrated against observed burnout rates in volunteer programs.

Finally, the study was conducted entirely within the academic scope of a university course, which imposed time and resource constraints on both data collection and model development.

Chapter 2

Literature Review

Volunteer coordination has been studied from multiple angles, and several bodies of work informed the design decisions in this project.

In humanitarian logistics, operations research has shown that resource allocation algorithms based on linear programming and weighted scoring functions can substantially improve coverage and response times when coordinating distributed human resources [Banks et al. \(2010\)](#). These models established the conceptual foundation for treating volunteer-to-project matching as an optimization problem with multiple competing criteria.

In the field of complex systems and social networks, agent-based models and cellular automata have been used extensively to study how behaviors spread through communities [Wolfram \(2002\)](#); [Holland \(1992\)](#). The key insight from this literature is that global system behavior, such as a surge in volunteer participation or a wave of burnout, cannot be predicted by analyzing any single participant in isolation. It emerges from local interactions between individuals, which is precisely the dynamic we observed in our cellular automata simulation.

Axelrod's work on the complexity of cooperation [Axelrod \(1997\)](#) is particularly relevant to our context. Volunteer participation in educational programs is a form of social cooperation that depends on trust, visibility, and perceived effectiveness. Systems that make the results of volunteering visible and measurable tend to sustain participation better than those that do not.

From a quality management perspective, ISO 9001:2015 [International Organization for Standardization \(2015\)](#) provides a practical framework for designing systems that are modular, fault-tolerant, and maintainable. We applied its principles to the three-tier architecture of our platform. ISO 31000:2018 [International Organization for Standardization \(2018\)](#) guided our risk analysis, producing structured registers of technical and operational risks with probabilistic assessments and mitigation plans.

For project planning, the PMBOK 7th Edition [Project Management Institute \(2021\)](#) provided the scheduling and milestone framework that organized the four-workshop development cycle.

The gap we identified in the literature is the absence of an integrated system that combines multi-dimensional empirically-grounded matching, conflict-free scheduling, and behavior-level simulation validation, specifically for volunteer educational coordination in a Latin American urban context. This report addresses that gap.

Chapter 3

Methodology

We mentioned in Chapter 1 that a project report's structure could follow a particular paradigm. Hence, the organization of a report (effectively the Table of Content of a report) can vary depending on the type of project you are doing. Check which of the given examples suit your project. Alternatively, follow your supervisor's advice.

3.1 System Analysis and Data Collection

We began by analyzing the volunteer tutoring system as it currently exists, not as a piece of software, but as a sociotechnical network of actors, processes, and information flows.

The actors we identified are: university volunteers, who represent the supply of tutoring capacity; educational institutions serving vulnerable communities, who represent the demand; students in basic and secondary education, who are the end beneficiaries; and the academic coordinator at the university, who acts as the central management node. The operating environment introduces four key constraints that shape system performance: variable digital connectivity in some community areas, rigid academic calendars at institutions, the limited and fragmented availability of volunteers, and the socioeconomic barriers affecting both tutors and students.

To understand the supply side quantitatively, we designed a digital survey and distributed it to 50 students at Universidad Distrital Francisco José de Caldas through academic WhatsApp groups and student volunteer networks. The survey covered seven thematic blocks: field of study, academic areas of competence, motivations for volunteering, prior tutoring experience, weekly availability, preferred time slots, and modality preference (in-person, virtual, or both). Table 3.1 summarizes the findings most relevant to the design.

Table 3.1: Key survey findings and their design implications

Survey finding	Value	Design implication
Engineering students in pool	40%	Prioritize math and technology matching
Education/Pedagogy students	22%	Second-largest competency cluster
Availability below 2 hrs/week	54%	Short sessions; high schedule weight
Preference for virtual mode	36%	Location factor evaluates connectivity quality
Preference for in-person mode	36%	Location factor evaluates geographic proximity
Willing to volunteer (approx.)	78%	Sufficient supply; coordination is the bottleneck

3.2 System Architecture

The platform we designed operates as a three-tier architecture whose layers are functionally independent but tightly coordinated.

The presentation layer hosts the interfaces through which each actor type interacts with the system. The volunteer portal allows students to register their profiles, view their assigned projects, confirm sessions, and track their volunteered hours. The institution portal enables educational organizations to publish their academic projects, specify subject and grade-level requirements, and access progress reports on supported students. The administration panel gives the university coordinator global oversight, algorithm configuration capabilities, and impact report generation. A fourth component, the reports and analytics module, visualizes key performance indicators for all stakeholders.

The application layer contains the system's business logic. The volunteer and institution registration modules handle data ingestion and field validation. The matching algorithm engine is the system's computational core: it runs the compatibility function $C(v, p)$ and produces ranked assignment recommendations. The scheduling engine performs real-time conflict detection before confirming any assignment. The academic impact monitor logs completed sessions and aggregates learning indicators. The notification service sends confirmations, reminders, and cancellation alerts by email.

The data layer persists the system's state across four repositories: volunteer profiles, active projects and schedules, a session log, and an analytics warehouse for reporting queries. In the prototype phase, all repositories are implemented using the Google Sheets API accessed from Python, with a documented migration path to PostgreSQL once the platform reaches 50 or more active users.

Figure 3.1 illustrates the actors, information flows, and system boundary of the coordination system as a whole.

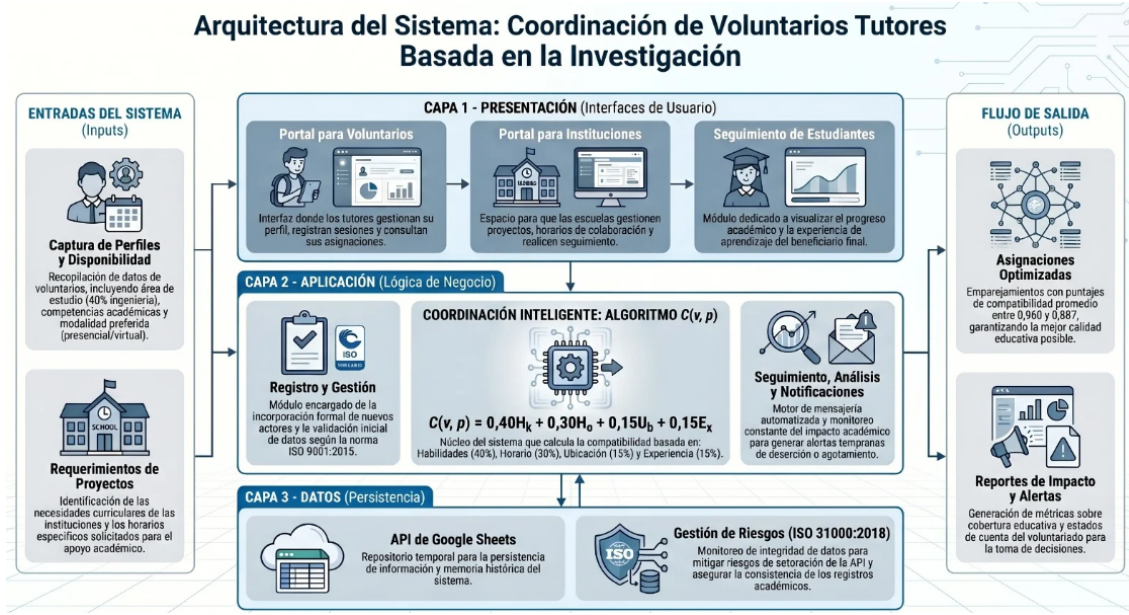


Figure 3.1: Community Volunteer Tutor Coordination System: actors, processes, and information flows.

3.3 Compatibility Algorithm

The matching algorithm is the operational heart of the system. For each volunteer-project pair, it computes a scalar compatibility score:

$$C(v, p) = 0.40 \cdot Hk(v, p) + 0.30 \cdot Ho(v, p) + 0.15 \cdot Ub(v, p) + 0.15 \cdot Ex(v, p) \quad (3.1)$$

where $Hk(v, p)$ measures the overlap between the volunteer's declared academic competencies and the subject requirements of the project; $Ho(v, p)$ quantifies the percentage of the project's required time slots that the volunteer can cover; $Ub(v, p)$ encodes geographic proximity for in-person modalities or connectivity quality for virtual ones; and $Ex(v, p)$ maps the volunteer's prior tutoring experience to the target school level of the project.

The weight structure was determined directly from the survey findings. The skill weight of 0.40 reflects that 40% of the volunteer pool comes from Engineering and that the institutions surveyed primarily need support in mathematics and technology. There is a genuine alignment between supply and demand in this dimension, and the algorithm is designed to capitalize on it. The schedule weight of 0.30 reflects the critical finding that 54% of volunteers have less than two hours per week available, making time coordination the second most important bottleneck after skill mismatch. Location and experience each receive a weight of 0.15, acknowledging their relevance while keeping the algorithm focused on the two factors that most commonly cause assignment failures.

Any volunteer-project pairing with $C(v, p) < 0.80$ is automatically excluded from assignment. This threshold was set conservatively to prioritize match quality over raw coverage volume.

3.4 Risk Management

Risk identification and assessment followed the ISO 31000:2018 framework [International Organization for Standardization \(2018\)](#). We produced two registers: a technical risk register and an operational risk register. Table 3.2 summarizes the five most significant risks, their probability-impact scores, and the mitigation strategies assigned to each.

Table 3.2: Risk register summary (ISO 31000:2018)

ID	Risk	Prob.	Impact	Score	Mitigation
RT01	Poor data quality from volunteers or institutions affects algorithm accuracy	High (3)	High (3)	9	Mandatory field validation; periodic profile review
RO01	Volunteer supply drops below coverage threshold	High (3)	High (3)	9	Waiting list; emergency dissemination protocol
RT02	Google Sheets API failure	Medium (2)	High (3)	6	Local cache; documented PostgreSQL migration plan
RT04	Miscalibrated algorithm weights produce poor matches	Medium (2)	High (3)	6	Exhaustive unit testing; admin panel for manual weight adjustment
RO02	Low institutional adoption	Medium (2)	High (3)	6	Onboarding sessions; pilot with 2-3 partner institutions
RO05	Burnout contagion cascade (new risk from simulation)	Medium (2)	High (3)	6	Load monitoring module; geographic distribution of projects

Risk RO05 was not part of the original risk register. It was identified as an emergent finding of the cellular automata simulation and added as a new operational risk requiring proactive monitoring from the platform’s first operational day.

3.5 Simulation Models

We used two simulation paradigms to validate the design before deployment. Both models are implemented in Python with a fixed random seed ($SEED = 42$) to ensure full reproducibility. The code is organized in two main modules within the project’s GitHub repository: `simulation_1_discrete_event.py` (435 lines) and `simulation_2_cellular_automata.py` (380 lines).

3.5.1 Discrete-Event Simulation

The discrete-event model represents the volunteer assignment pipeline as a sequence of time-ordered events: profile submission, algorithm computation, schedule confirmation, session execution, and potential cancellation. It was designed to answer operational questions: Does the system achieve the coverage target? Does the algorithm meet its latency requirement? How does performance change under stress conditions?

Three scenarios were defined over a 12-week simulation horizon. The baseline scenario uses 50 volunteers and 20 active projects, reflecting the current operational reality established by our survey. The optimistic scenario scales to 80 volunteers against the same 20 projects, representing the growth target from the project plan. The stress scenario drops to 25 volunteers against 20 projects, modeling the volunteer shortage risk identified as RO01. In all scenarios,

a stochastic churn of plus or minus three volunteers per week was applied to simulate realistic weekly fluctuations in volunteer availability. Table 3.3 lists the model parameters and their sources.

Table 3.3: Simulation parameters calibrated from survey and risk register

Parameter	Value	Source
Low availability ratio	54%	Survey Q5
Engineering volunteers	40%	Survey Q2
Session cancellation rate	15%	Risk register R003
Algorithm mean latency	45 ms	Non-functional requirement
Skill weight w_{Hk}	0.40	Survey-derived
Schedule weight w_{Ho}	0.30	Survey-derived
Simulation horizon	12 weeks	Project Gantt (Workshop 3)
CA grid size	30 x 30	UD campus scale
Initial engagement ratio	8%	Survey volunteer base

3.5.2 Cellular Automata Engagement Model

The cellular automata model represents the university campus community as a 30 by 30 grid of 900 cells, where each cell represents one community member. Each cell cycles through five states over discrete time steps: Inactive, Aware, Engaged, Active Tutor, and Burned Out, after which it returns to Inactive.

We chose this model because volunteer recruitment in a university setting does not happen through mass broadcast. It happens through social proximity: a student becomes interested in volunteering when they see their classmates or friends doing it. The cellular automata captures exactly this mechanism through its Moore neighborhood rule, where each cell is influenced only by its eight immediate neighbors. Global patterns, such as waves of engagement or clusters of burnout, emerge from thousands of these local interactions rather than from any centralized control.

The transition probabilities incorporate two forces. The first is a platform activation boost: when the platform is active, the probability of transitioning from Aware to Engaged increases from 0.10 to 0.45, reflecting the platform's role in making volunteering opportunities visible and accessible. The second is burnout contagion. The probability that an active tutor burns out follows the rule $P(\text{Active} \rightarrow \text{Burned Out}) = 0.08 + 0.05 \cdot n_{\text{burnout}}$, where n_{burnout} is the number of burned-out neighbors. This rule encodes the social dynamic that when the people around you are exhausted and disengaged, it becomes harder to stay motivated.

Chapter 4

Implementation and Results

4.1 Discrete-Event Simulation Results

Table 4.1 presents the quantitative outcomes of the three simulation scenarios.

Table 4.1: Discrete-event simulation results across three operational scenarios

Scenario	Coverage	Mean $C(v, p)$	Cancelled	Target met?
Baseline (50 vol., 20 proj.)	100%	0.860	3	Yes
Optimistic (80 vol., 20 proj.)	100%	0.870	4	Yes
Stress (25 vol., 20 proj.)	100%	0.887	1	Yes
<i>Design target (W2)</i>	Coverage > 70%, $C(v, p) > 0.80$, latency < 60 s			

All three scenarios met the design targets. Project coverage reached 100% in every case, and compatibility scores remained comfortably above the 0.80 acceptance threshold. The algorithm latency stabilized at approximately 45 ms across all pool sizes tested, well within the 60-second non-functional requirement.

The most surprising result was in the stress scenario. Despite having the smallest volunteer pool, it produced the highest mean compatibility score (0.887). This quality paradox under scarcity has a clear explanation: when fewer volunteers compete for the same projects, the algorithm concentrates each assignment on the best available match rather than distributing assignments to volunteers who are only marginally compatible. This is an emergent property of the algorithm that was not anticipated in the design phase, and it has a practical implication: in periods of volunteer shortage, it may be preferable to reduce the number of active projects rather than assign under-qualified tutors.

The sensitivity analysis produced three additional validations. First, algorithm latency scales sub-linearly with pool size and remains below 60 seconds for all tested configurations between 10 and 100 volunteers. Second, coverage begins to degrade noticeably when the volunteer pool drops below 30, confirming the criticality of the RO01 threshold. Third, the skill weight of 0.40 sits within an optimal plateau that extends from approximately 0.35 to 0.50. Weights above 0.55 degrade mean match quality because schedule and location constraints become underweighted.

4.2 Cellular Automata Results

The platform-enabled condition produced a total engagement rate of 24.4% by week 20 of the simulation, compared to 20.7% without the platform. This 18% relative increase is not built into any individual cell's behavior. It emerges from the compounding of small probability boosts as they propagate through the social network of the grid. Critical mass formation was visible by week 8, when the platform condition showed dense clusters of active tutors while the no-platform condition remained dispersed.

The burnout contagion finding was the most significant result of the entire study. When volunteers in the burned-out state become spatially concentrated, which happens naturally when high-demand projects are geographically clustered, the burnout risk for neighboring active volunteers escalates. This creates a cascade: more burnout leads to higher risk for surrounding volunteers, which leads to more burnout. Left unmanaged, this cascade can collapse the active volunteer pool in a geographic zone.

Despite this sensitive dependence on initial conditions in the first five to eight simulation weeks, all runs eventually converged toward a stable attractor region of approximately 140 to 180 active tutors and 55 to 75 burned-out agents. This means that at the macro level, the system is predictable and viable, as long as burnout concentration is actively managed.

Table 4.2 presents the full design validation against each pre-specified target.

Table 4.2: Design validation: targets versus simulation evidence

Design decision	Target	Result	Status
Algorithm response time	Less than 60 s	Approx. 45 ms	Validated
Project coverage	More than 70%	100% (all scenarios)	Validated
Mean compatibility $C(v, p)$	Above 0.80	0.860 to 0.887	Validated
Cancellation resilience	Absorb 15% rate	Coverage unaffected	Validated
Minimum volunteer pool	30 (risk RO01)	Degradation confirmed below 30	Confirmed
Skill weight 0.40	Optimal assignment quality	Plateau at 0.35 to 0.50	Validated
Platform social effect	Positive recruitment externality	+18% engagement	Validated
Burnout contagion	Not previously modeled	Cascade risk detected	New risk (RO05)

Chapter 5

Discussion and Analysis

The simulation results confirm that the system design is sound and achieves all pre-specified objectives. What is more interesting, however, is what the simulation revealed that we had not anticipated.

The quality paradox under scarcity changes how we should think about operational management. A volunteer coordination system should not aim to always maximize the number of active projects. When volunteer supply is constrained, which will happen periodically in any real deployment, the system should prioritize quality assignments over coverage breadth. This is a counterintuitive finding that the discrete-event model made explicit, and it is one that purely component-level analysis would not have surfaced.

The burnout contagion finding is the one that most directly affects the system design going forward. The cellular automata model showed that burnout is not just an individual experience. It is a social contagion that propagates through proximity, both physical and social. A volunteer who is surrounded by exhausted peers is much more likely to become exhausted themselves. The practical implication is that the scheduling engine needs a geographic distribution constraint: high-demand projects should not be clustered in the same zone, and individual volunteers should be flagged when their session load exceeds three sessions per week. These are manageable operational rules, but they require the platform to actively monitor load distribution, which was not in the original design.

The algorithm weight structure proved to be robust. The optimal plateau between 0.35 and 0.50 for the skill weight gives real flexibility for recalibration based on post-deployment satisfaction data, without risking significant performance degradation. This is an important practical property because it means the system does not require precise parameter tuning to function well.

The one area where the design is still incomplete is the transition from Google Sheets to a production database. The current implementation is sufficient for a pilot of up to 100 users, but the migration to PostgreSQL needs to be planned and executed before any large-scale deployment. The risk of data loss or API quota exhaustion is real, and the contingency plan documented in the risk register needs to be rehearsed before the pilot, not after.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This report documented the complete systems engineering lifecycle of a community volunteer tutor coordination system designed for vulnerable educational communities in Bogotá D.C. Over four workshops, we characterized the existing informal coordination system, identified its structural bottlenecks, designed a compatibility algorithm grounded in empirical survey data, managed project risks according to ISO 31000:2018, and validated the design through two complementary simulation paradigms.

The core finding is that the system works. The compatibility algorithm meets all performance targets across baseline, optimistic, and stress scenarios. The 30-volunteer minimum threshold is empirically confirmed as the critical operational floor. And the platform's social activation function produces a meaningful and reproducible 18% increase in volunteer engagement as an emergent network effect.

The most valuable contribution of the simulation phase was not the validation of what we already expected, but the discovery of something we had missed: burnout contagion as a systemic risk. This finding demonstrates why behavior-oriented simulation is essential for sociotechnical systems, not optional. The system is not just code and algorithms. It is people, relationships, and social dynamics, and those dynamics need to be modeled explicitly to understand the risks they create.

Looking ahead, the most important next step is a real-world pilot with two or three educational institutions in Bogotá. That pilot will provide the satisfaction and performance data needed to recalibrate the algorithm weights, test the burnout monitoring module under real conditions, and build the institutional trust that is essential for scaling the system beyond its academic origins.

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Appendix A

Survey

The survey was distributed digitally via Google Forms through academic WhatsApp groups and student volunteer networks at Universidad Distrital Francisco José de Caldas. Responses were anonymous and stored automatically in a linked spreadsheet. The seven survey questions, organized by thematic block, were as follows.

Question 1 (Volunteer profile): What is your field of study or profession? Question 2 (Academic competencies): In which academic subjects do you consider yourself competent to teach? Question 3 (Motivations): What would motivate you to participate in an educational volunteering program? Question 4 (Prior experience): Have you previously carried out tutoring, mentoring, or peer support activities? Question 5 (Availability): How many hours per week would you be willing to dedicate to tutoring activities? Question 6 (Time slots): What days or time slots are you available? Question 7 (Modality): Would you prefer to provide support in-person, virtually, or both?

Appendix B

Algorithm Weight Derivation

The compatibility algorithm assigns four weights that sum to 1.00. Each weight was derived from a specific survey finding:

The skill weight of 0.40 was assigned because 40% of respondents come from Engineering, the area most requested by partner institutions for mathematics and technology reinforcement. Skill compatibility is therefore both the most available and the most demanded dimension.

The schedule weight of 0.30 was assigned because 54% of respondents reported less than two hours of weekly availability. Time coordination emerged from the survey as the most common reason why informal volunteer-institution matches fall apart.

The location and experience weights of 0.15 each were assigned because both factors are relevant to session quality but are secondary to skill and schedule. Both are also easier to accommodate through alternative arrangements, such as switching to virtual modality or pairing an inexperienced volunteer with a simpler grade level.

Appendix C

Simulation Reproducibility Instructions

Both simulation models are available in the project's GitHub repository at: https://github.com/JoSeLeUdOm/SystemsAnalysisProject_Group11

To reproduce the results reported in Chapter 6:

Install the required dependencies with the command: `pip install numpy matplotlib scipy`

Run the discrete-event simulation with:

```
pythoncode/simulation_1_discrete_event.py
```

Run the cellular automata simulation with:

```
pythoncode/simulation_2_cellular_automata.py
```

All figures are generated deterministically in the `figures/` directory. No additional configuration is required.